

DAY 11

THE BLACKJACK CAPSTONE PROJECT

WHAT WE WILL MAKE BY THE END OF THE DAY

- This is our first Capstone Project, **A capstone project** is a big, final assignment—usually done at the end of a degree, diploma, or certification program—that’s meant to bring together and apply everything you’ve learned in that course of study. Think of it like your “final boss battle” in school: instead of just taking an exam, you create something that proves you can use your skills in a real-world way.
- This would be the hardest thing i would’ve built so far, **so it’s adviceable to revise my past notes before this point** a
- [free online blackjack game](#)
- [a video on blackjack](#)
- [blackjack basic strategy](#)
- [tips on memorizing basic strategy](#)
- [final project demo](#)

BLACKJACK PROGRAM REQUIREMENTS AND GAME RULES

- Blackjack is also known as 21
- *Yes — the “must hit on 17” rule in blackjack applies **only to the dealer**, not to players.*

Here’s how it works:

- *In most casinos, the dealer must **keep drawing cards (“hit”)** until their hand totals at least **17**.*
- *If the dealer reaches 17 or higher, they must **stand** (stop taking cards).*
- *In some variations, the dealer must hit on a “**soft 17**” (a 17 that includes an Ace counted as 11, like Ace + 6), but always stand on a **hard 17** (no Ace counted as 11).*
- *Players, on the other hand, are **never forced** to hit or stand at a certain number — they can decide freely.*

This rule exists so that the dealer’s play is completely automatic and consistent — no strategy or choice, just fixed behavior.

- There are hints in the coding file, use as necessary, remember to research, run the solution, build on your own process using pseudocode and a flow chart, create your todo’s and then try building your own

BLACKJACK CAPSTONE PROJECT

- [flowcharts](#) are more than just plan drafts, but more of a documentation on the stable code logic
- A function shouldn't always be called multiple times, you can store a specific output of a function you might need in a variable and that could help you prevent different values, bulky code and easier logic
- if you don't return an iterable value after you define your function, if you use sum on it it'd refer to your function, not a list
- i couldn't make ace hold two values and change it depending on the scenario, besides that felt a little more complex, so i made a variable called ace that held 11 and changed it to 1 later down the code depending on certain conditions
- To make a project restart after a loop, always remember to terminate the loop before using recursion
- [changing the content of a list using the for loop didn't work](#) ,

Hint

- you can use an underscore '_' as a temporary variable in a for loop where the temporary variable doesn't really matter
- you can use the triple quotation for infunction comments
- it's not that the += operator doesn't work on list, but rather you can only use it on a list and another list, it kinda serves like the extend function, so if it's just a single item and not a list use the append function
- when running condition for blackjack i forgot to add an extra condition for the length of cards to be only 2, as you can have a normal 21 or a blackjack
- she she created a condition for a blackjack and equalled it to the score to a 0, to differentiate from a normal 21,
- you can only call a function after it's been declared
- there is a list function called list.remove(), you can use it conjunction with the append function to replace a certain item on a list, if only the order doesn't matter
- Although i was able to make it work, my code has some better ideas, but a whole buttload of inefficiencies, so i should work on it overtime as I'm moving on
- n Python, I can use parentheses together with logical operators (and, or, not) to chain multiple conditions in a single if statement.
 - if (sum(fruit) == 21 and len(fruit) == 2) and (sum(grains) == 21 and len(grains) == 2):
print("yay")